Supplementary Materials for “FMDI: Frequency-Shaped Flow Matching for Incomplete Time-Series Data Repair”

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I. Overview

This supplementary material provides derivations, extended experimental details, additional result tables, and additional figures for the FMDI manuscript. The current version includes the derivation of the frequency-shaped flow matching path. Additional sections can be added as the submission package is finalized.

II. Supplementary Derivation for FMDI

This file records the derivation behind the frequency-shaped flow matching path used by FMDI. It is intended as supplemental artifact material rather than an appendix inserted into the ICDE main paper.

A. From Time-Domain Flow Matching to the RFFT Domain

Let $\mathbf{x}_1 \sim p_{\text{data}}$ be a data sample and $\mathbf{z} \sim p_{\text{base}}$ be a base noise sample. A general flow matching path can be written as

$$\mathbf{x}_t = a(t)\mathbf{x}_1 + b(t)\mathbf{z}, \qquad t \in [0, 1], \tag{S1}$$

with $a(0) = 0$, $b(0) = 1$, $a(1) = 1$, and $b(1) = 0$. The path velocity is

$$\mathbf{u}_t = \frac{d\mathbf{x}_t}{dt} = \dot{a}(t)\mathbf{x}_1 + \dot{b}(t)\mathbf{z}. \tag{S2}$$

Let $\mathcal{F}$ be the orthonormal RFFT along the temporal dimension. Since $\mathcal{F}$ is linear,

$$\mathbf{X}_t = \mathcal{F}(\mathbf{x}_t) = a(t)\mathbf{X}_1 + b(t)\mathbf{Z}, \tag{S3}$$

and

$$\mathbf{U}_t = \mathcal{F}(\mathbf{u}_t) = \dot{a}(t)\mathbf{X}_1 + \dot{b}(t)\mathbf{Z}. \tag{S4}$$

Thus, flow matching can be equivalently described in the RFFT domain, and a time-domain velocity network corresponds to a frequency-domain velocity after applying $\mathcal{F}$ to its output.

B. Frequency-Shaped Path

FMDI replaces the uniform frequency-domain base noise with

$$\mathbf{Z}_s^{(f)} = s_f \mathbf{Z}^{(f)}, \qquad s_f > 0. \tag{S5}$$

The general shaped path becomes

$$\mathbf{X}_t^{(f)} = a(t)\mathbf{X}_1^{(f)} + b(t)s_f\mathbf{Z}^{(f)}. \tag{S6}$$

For fixed $\mathbf{X}_1$, the conditional variance of frequency bin f is proportional to $b^2(t)s_f^2$. Therefore, the path keeps the flow-matching endpoint structure but makes the uncertainty schedule frequency dependent.

For the linear path $a(t) = t$ and $b(t) = 1 - t$,

$$\mathbf{X}_t^{(f)} = t\mathbf{X}_1^{(f)} + (1-t)s_f\mathbf{Z}^{(f)}, \tag{S7}$$

and the target velocity is

$$\mathbf{U}^{(f)} = \mathbf{X}_1^{(f)} - s_f\mathbf{Z}^{(f)}. \tag{S8}$$

Because $\mathbf{X}_t^{(f)} = t\mathbf{X}_1^{(f)} + (1-t)s_f\mathbf{Z}^{(f)}$, the same velocity can also be written as

$$\mathbf{U}^{(f)} = \frac{\mathbf{X}_1^{(f)} - \mathbf{X}_t^{(f)}}{1-t}, \qquad t < 1. \tag{S9}$$

The algebraic form of the linear velocity is the same as ordinary linear flow matching, but the state distribution changes because $\mathbf{X}_t^{(f)}$ contains the shaped noise term $s_f\mathbf{Z}^{(f)}$.

C. Parseval Calibration

For a real signal of length L, the orthonormal RFFT satisfies the half-spectrum Parseval identity

$$\|\mathbf{x}\|_2^2 = \sum_{f=0}^{L_f-1} \omega_f |\mathbf{X}^{(f)}|^2, \tag{S10}$$

where $\omega_f = 1$ for the DC bin and the Nyquist bin, if present, and $\omega_f = 2$ for interior bins.

The shaped base noise has total expected RFFT-domain energy proportional to

$$\sum_{f=0}^{L_f-1} \omega_f s_f^2. \tag{S11}$$

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To make shaping redistribute energy without changing the total energy scale, FMDI imposes

$$\sum_{f=0}^{L_f-1} \omega_f s_f^2 = L. \tag{S12}$$

Given any positive unnormalized profile $\widetilde{s}_f^2$, the calibrated scale is

$$s_f^2 = \frac{L\widetilde{s}_f^2}{\sum_{j=0}^{L_f-1} \omega_j \widetilde{s}_j^2}. \tag{S13}$$

D. Training Objective

Let $\widehat{\mathbf{u}}_\theta$ be the time-domain velocity predicted by the network and $\mathbf{u} = \mathcal{F}^{-1}(\mathbf{U})$ be the target velocity. For conditional repair, only entries hidden from the conditioning context but available for supervision are used in the primary loss:

$$\mathcal{L}_{\text{time}} = \frac{\|\mathbf{r} \odot (\widehat{\mathbf{u}}_\theta - \mathbf{u})\|_2^2}{\sum_{k,l} r_{k,l}}. \tag{S14}$$

The frequency-domain residual is

$$\widehat{\mathbf{U}}_\theta - \mathbf{U} = \mathcal{F}(\widehat{\mathbf{u}}_\theta) - \mathbf{U}. \tag{S15}$$

Using the Parseval weights, FMDI adds

$$\mathcal{L}_{\text{freq}} = \frac{1}{L}\mathbb{E}_{B,K} \sum_{f=0}^{L_f-1} \omega_f |\widehat{\mathbf{U}}_\theta^{(f)} - \mathbf{U}^{(f)}|^2. \tag{S16}$$

The full objective is

$$\mathcal{L} = \mathcal{L}_{\text{time}} + \lambda_{\text{freq}}\mathcal{L}_{\text{freq}}. \tag{S17}$$

E. Sampling ODE and Relation to FSDI

FMDI samples from

$$\mathbf{X}_{t=0}^{(f)} = s_f \mathbf{Z}^{(f)} \tag{S18}$$

and solves

$$\frac{d\mathbf{X}_t^{(f)}}{dt} = \mathbf{V}_\theta^{(f)}(\mathbf{X}_t, t, \mathbf{c}), \qquad \mathbf{V}_\theta = \mathcal{F}(v_\theta). \tag{S19}$$

With explicit Euler steps, this becomes

$$\mathbf{X}_{t+\Delta t}^{(f)} = \mathbf{X}_t^{(f)} + \Delta t\, \mathbf{V}_\theta^{(f)}(\mathbf{X}_t, t, \mathbf{c}). \tag{S20}$$

FSDI-style frequency-shaped diffusion uses a discrete marginal of the form

$$\mathbf{X}_t^{(f)} = \sqrt{\bar{\alpha}_t}\mathbf{X}_1^{(f)} + \sqrt{1-\bar{\alpha}_t}\, s_f \mathbf{Z}_t^{(f)}. \tag{S21}$$

FMDI keeps the frequency-shaped uncertainty idea but replaces the reverse Markov posterior and noise-prediction target with a continuous velocity field and ODE integration.

III. Additional Experimental Details

This section will contain extended dataset, implementation, hyperparameter, and runtime-profiling details that are too detailed for the main paper.

IV. Additional Tables and Figures

This section provides supplementary qualitative repair examples over all channels for representative saved FMDI runs.

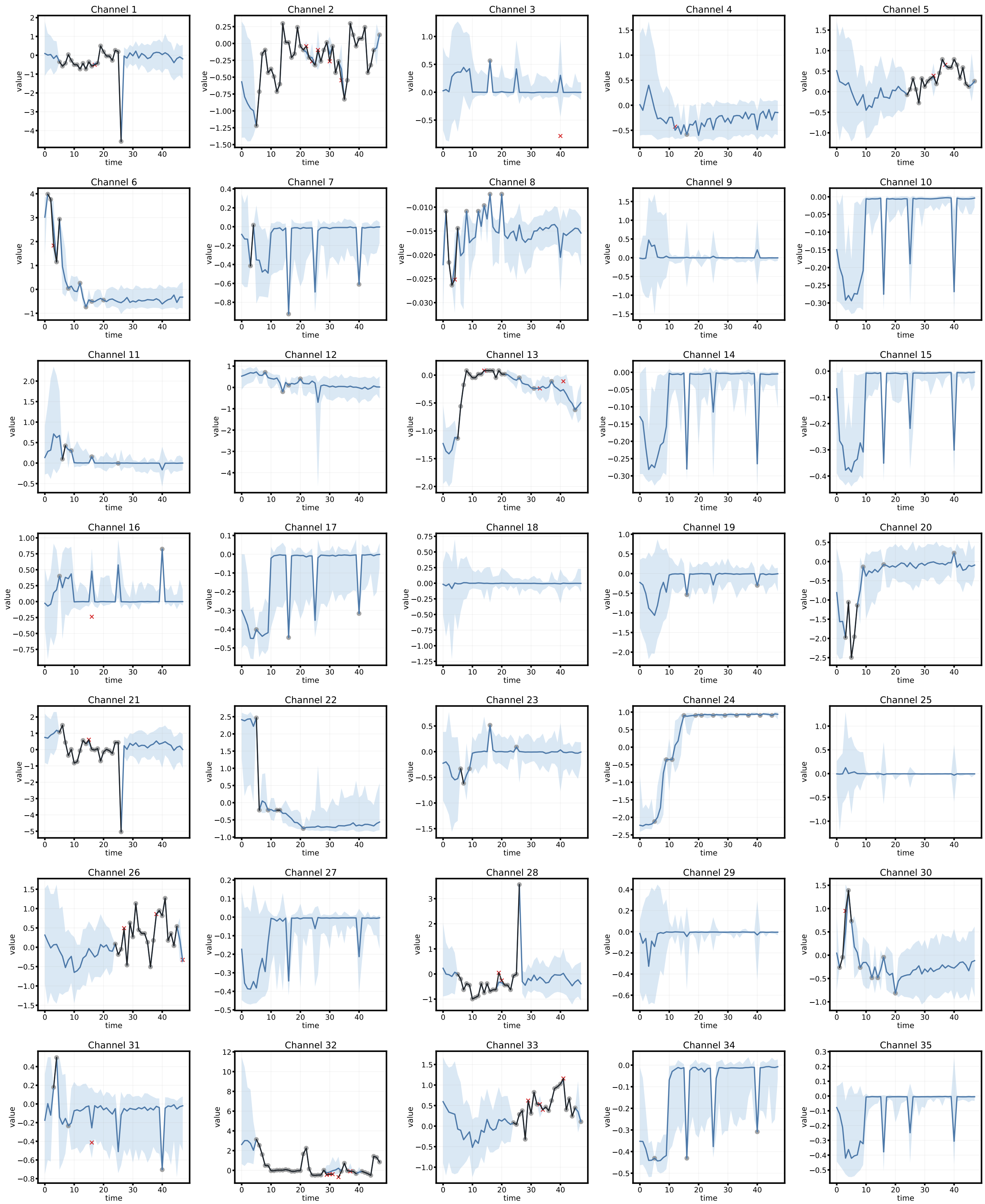

Fig. S1. Visualization of imputation results on Physionet2012 dataset from Channel 1 to Channel 35 with 10% missing. The blue solid line represents the sample median, the blue shaded region represents the 10–90% sample interval, the black line and gray markers represent available ground-truth entries, and the red crosses represent held-out target entries.

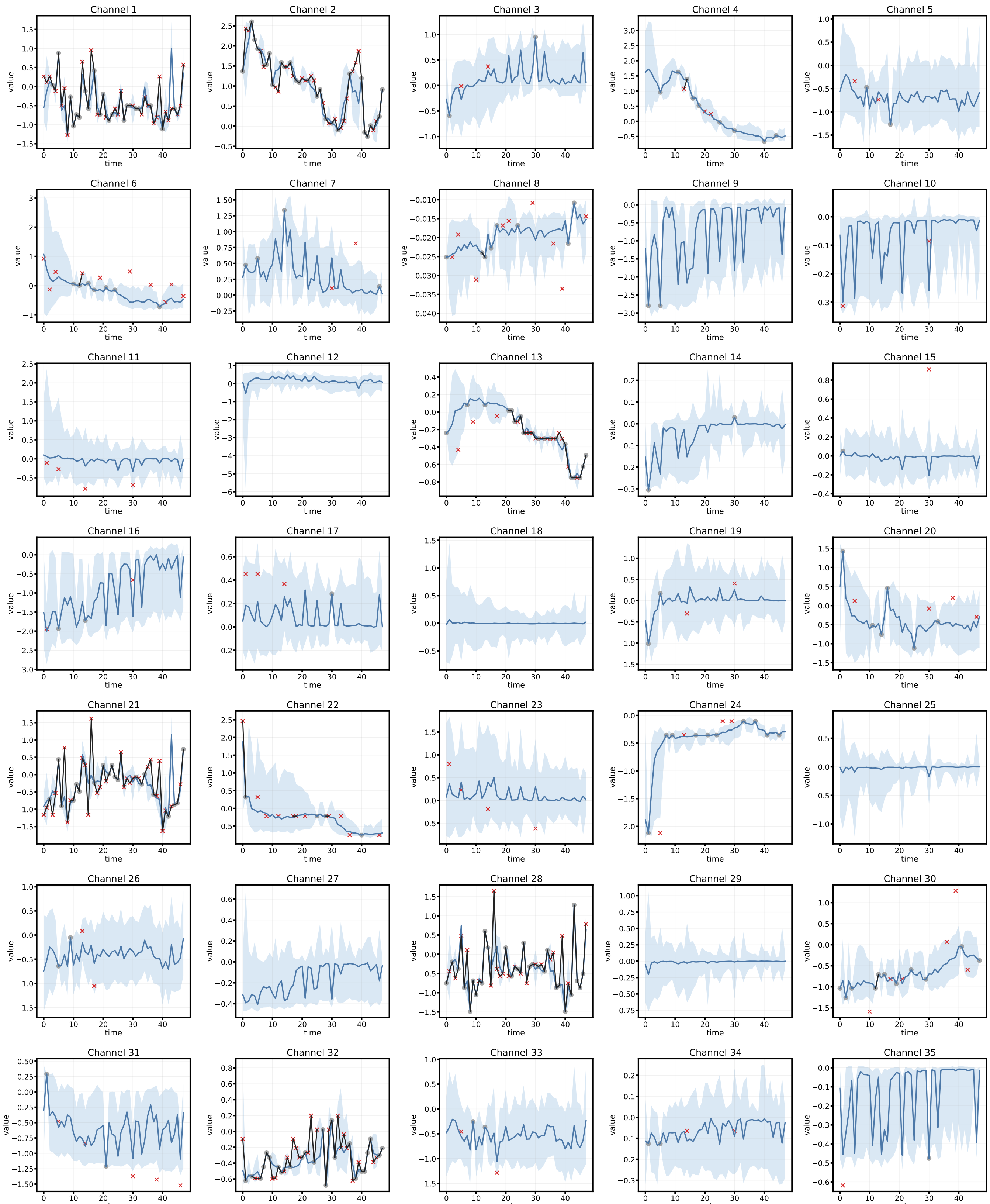

Fig. S2. Visualization of imputation results on Physionet2012 dataset from Channel 1 to Channel 35 with 50% missing. The blue solid line represents the sample median, the blue shaded region represents the 10–90% sample interval, the black line and gray markers represent available ground-truth entries, and the red crosses represent held-out target entries.

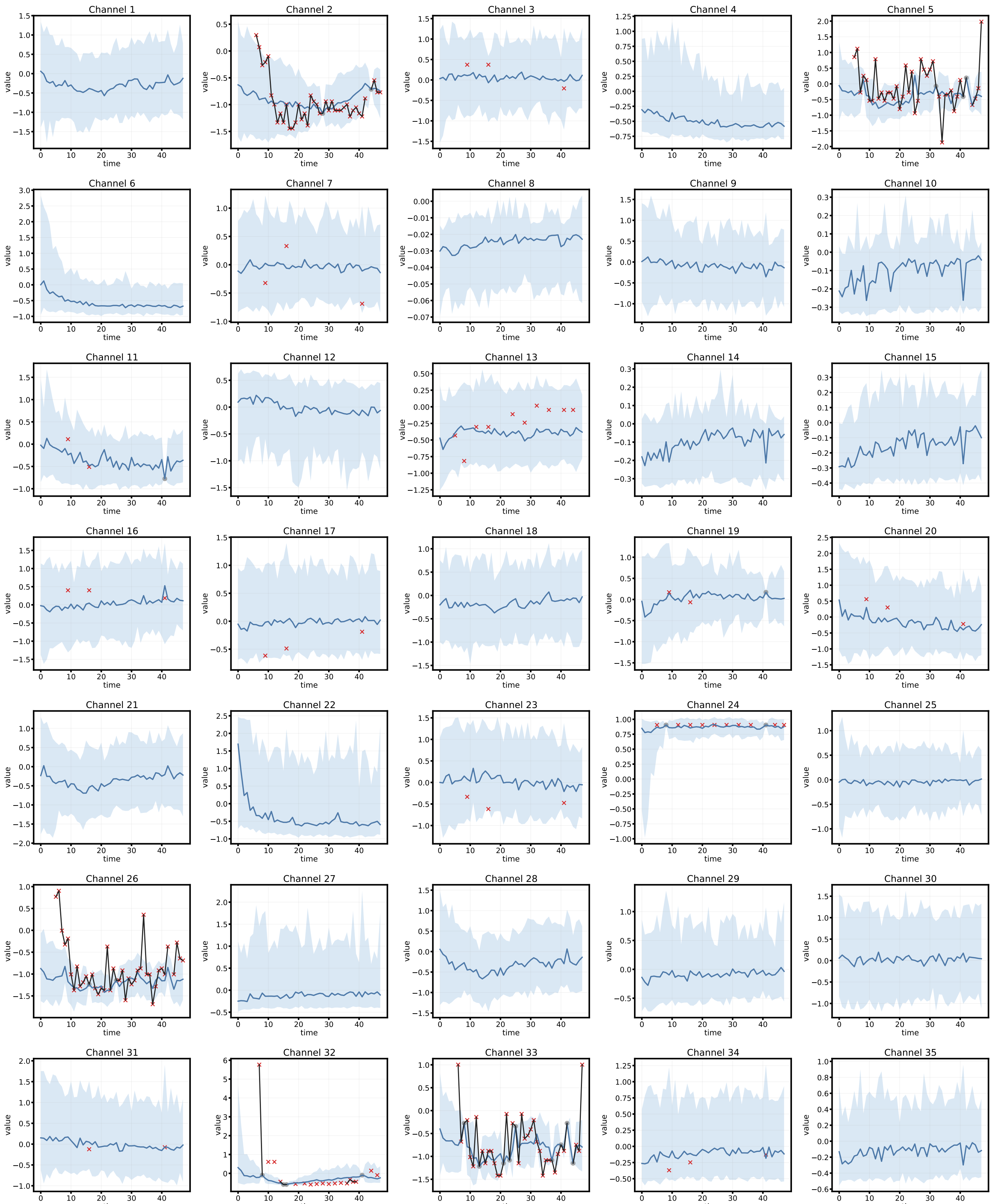

Fig. S3. Visualization of imputation results on Physionet2012 dataset from Channel 1 to Channel 35 with 90% missing. The blue solid line represents the sample median, the blue shaded region represents the 10–90% sample interval, the black line and gray markers represent available ground-truth entries, and the red crosses represent held-out target entries.

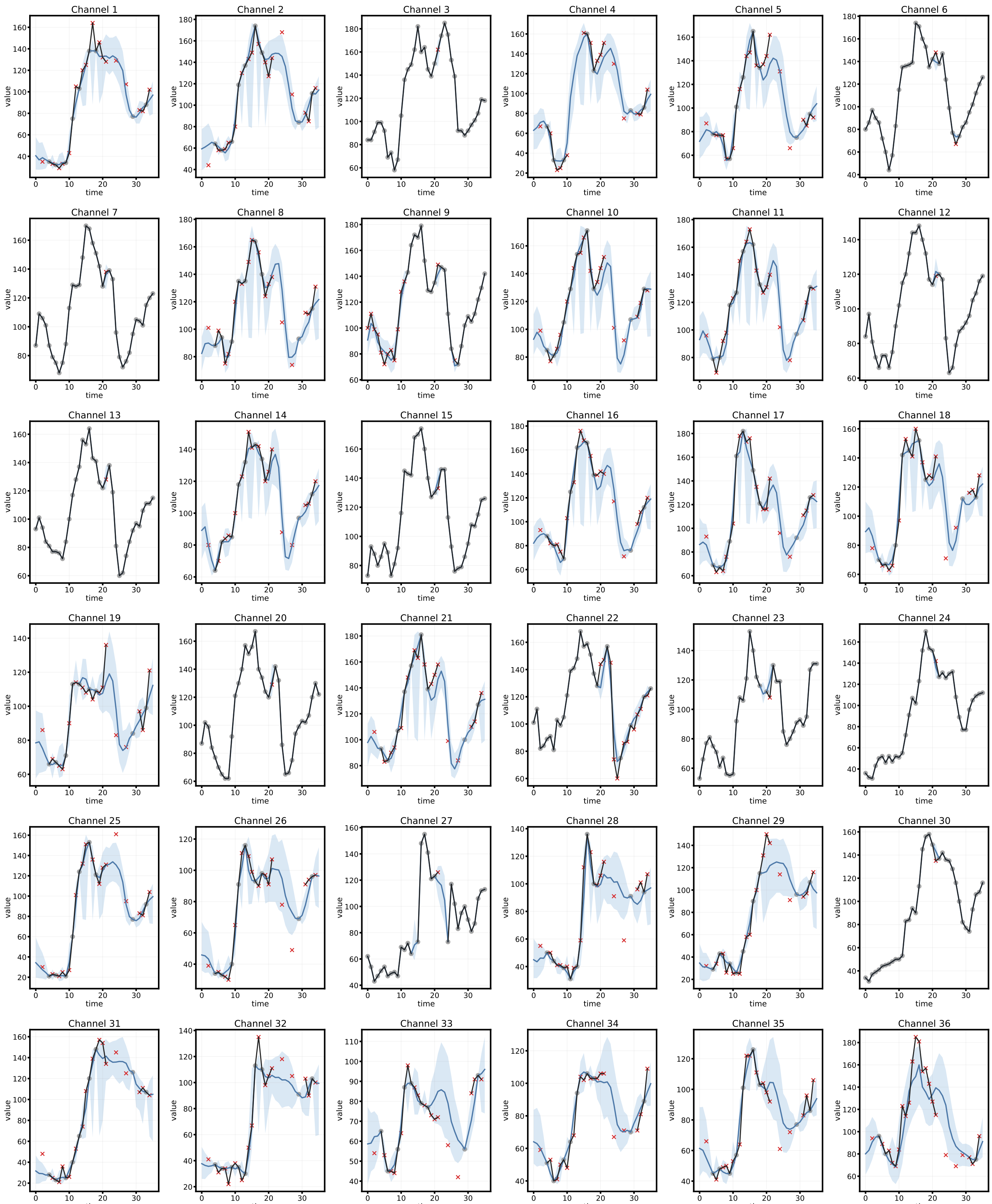

Fig. S4. Visualization of imputation results on AQI dataset from Channel 1 to Channel 36. The blue solid line represents the sample median, the blue shaded region represents the 10–90% sample interval, the black line and gray markers represent available ground-truth entries, and the red crosses represent held-out target entries.